

Peijie (Max) Ma

✉ pm3688@nyu.edu ☎ +1 (516) 288-0844 in [LinkedIn](#) ⚙️ [Tau Beta Pi \(Honor Society\)](#) G [Google Scholar \(148+ citations\)](#)

Education

New York University

Sept. 2023 - June 2027

- **B.S. Double Major** in Computer Engineering and Math **GPA: 4.0/4.0 (Dean's List)**
- **Coursework:** Embedded, Object Oriented Programming, Signal and System, Digital Logic, Computer Architecture, Electronics, Probability, Machine Learning, Data Structure and Algorithm, Linear Algebra
- **Honors & Awards:** The Myron M. Rosenthal Merit Award, awarded to the highest academic standing juniors

Projects

Comprehensive FR3 Radio Propagation Measurement and Modeling

Nov. 2023 - Present

Supervisor: [Prof. Theodore S. Rappaport](#) [🔗](#) (Founder of NYU WIRELESS)

- Conducted the world's first comprehensive upper mid-band measurement campaign at 6.75 and 16.95 GHz.
- Assisted in developing a detailed statistical channel model over 100 TX-RX location pairs.
- Analyzed and processed over 150 GB of power delay profiles, to form a unified data format for sharing.

AMAR: Automated Material-Aware 3D Reconstruction

Sept. 2023 - Present

Supervisors: [Prof. David Fouhey](#) [🔗](#) & [Prof. Theodore S. Rappaport](#) [🔗](#)

- Developed the ML-enabled method to detect multiple planes/tiny objects, achieving detection accuracy within 10 cm and classification precision exceeding 89%.
- Validated and calibrated the NYURay ray tracer using 30 different objects with 10 materials.
- Reconstructed an indoor environment of 2000 m² and an outdoor environment of 2 km² using Blender.

Experience

Test Engineer *Mini-Circuits*

May. 2025 - Sept. 2025

- Analyze and review with the manager RF, DC, and control schematics for proper functionality and stress derating using ANSYS, Draftsight, and ADS.
- Participate in preparing E-Run final design and H-Run specs from the customers over 20 products.
- Proficient in using spectrum-analyzer to measure S-parameters, noise-figure, Psat, P1dB, and stability, etc.

Teaching and Research Assistant *Intro to ECE 1002*

Feb. 2025 - Present

- Assisted in authoring a comprehensive textbook about the historical evolution, fundamental principles, and modern advancements of Electrical & Computer Engineering and Computer Science.
- Designed and published 13 weekly assignments aligned with lecture content together with detailed solutions.
- Provided academic mentorship to 160 students through interactive and personalized support.

Algorithm Engineer *Matrixtime Robotics Co. Ltd, Shanghai, China*

Dec. 2023 - Feb. 2024

- Standardized functions for a semiconductor defect inspection program, increasing accuracy by 92%.
- Optimized data acquisition algorithms, reducing processing time by 42% and improving chip yield detection efficiency by 85.1%.

Selected Publications

- [1] M. Ying, **P. Ma**, D. Shakya, T.S. Rappaport, "Multi-Stage Location Optimization Through Power Delay Profile Alignment Using Site-Specific Wireless Ray Tracing"(accepted). [\[PDF\]](#) [🔗](#)
- [2] M. Ying, D. Shakya, T. S. Rappaport, **P. Ma**, "Comprehensive FR1 (C) and FR3 lower and upper mid-band propagation and material penetration loss measurements and channel models in indoor environment for 5G and 6G" in *IEEE Open Journal of the Communications Society* (accepted). [\[PDF\]](#) [🔗](#)

Key Skills

Programming Languages: C++/C, Java, Verilog, Python, SQL, MatLab, HTML, Markdown, Latex, linux

Languages: English (Proficient), Chinese (Native)

Tools: Blender (Proficient), AutoCAD, Fusion 360, Microsoft Suite, Ansys, ADS, Spectrum Analyzer

Research Areas: Wireless Channel Modeling, Ray-Tracing, 3D Reconstruction, Computer Vision